

Paper Name: Cloud Computing Technologies (IOT3201)
Assignment

Group – B

1. Differentiate between IaaS, PaaS, SaaS, and serverless computing.
2. Describe cloud cube model.
3. Define cloud computing.
4. Discuss the economic benefits of cloud computing and evaluate hidden costs.
5. Analyze how cloud computing enhances business continuity and disaster recovery.
6. Evaluate the role of cloud computing in enabling global collaboration.
7. How is cloud cube model significant in cloud security?
8. Explain the public cloud deployment model with advantages and disadvantages.
9. What is EDA?
10. Critically examine security and privacy concerns in cloud computing environments.
11. Analyze the challenges of data portability and interoperability in multi-cloud strategies.
12. How EDA can support real-time applications? Give an example.
13. What is multi-cloud computing?
14. Describe the role of SOA.
15. Examine how cloud computing supports digital transformation in enterprises with practical scenarios.
16. Analyze the impact of multi-tenancy on performance isolation and data security.
17. Discuss the role of Service Level Agreements (SLAs) in cloud computing and evaluate their limitations.
18. Evaluate the importance of fault tolerance and high availability in cloud architecture.
19. Analyze how edge computing complements cloud computing in latency-sensitive applications.
20. Explain the NIST cloud computing reference architecture and its significance.
21. Explain key features of serverless computing.
22. What is cloud virtualization?
23. Identify the benefits and limitations of hybrid cloud.
24. How do the management and security play a crucial role in cloud computing architecture?
25. How does the community cloud model fulfil customers' requirements?
26. Analyze how the "perimeterized vs de-perimeterized" dimension affects cybersecurity strategies.
27. Evaluate the limitations of the Cloud Cube Model in hybrid and multi-cloud environments.
28. Discuss how data location (internal vs external) impacts compliance using the Cloud Cube Model.
29. Explain the key characteristics of cloud computing with examples.
30. Differentiate between cloud auditor and cloud broker.
31. What are the three key questions a customer should ask a cloud provider according to the Jericho Forum?
32. Evaluate the feasibility of using a Public Cloud for highly sensitive government applications.
33. Analyze cost-benefit trade-offs between Private Cloud and hybrid cloud for medium-scale enterprises.
34. Discuss architectural challenges in implementing Hybrid Cloud solutions.
35. Evaluate how vendor lock-in affects hybrid cloud strategies.
36. Analyze how community clouds can address industry-specific compliance requirements.
37. A university IT department maintains separate physical servers for different services: 1 server for the student portal, 1 server for the online learning system, 1 server for the email service, 1 server for the library database. Most of the time, these servers use only 25-40% of their capacity, This leads to high maintenance costs, increased electricity usage, and inefficient resource utilization. How can cloud virtualization help solve this problem, including facilities of cloud like auto-scaling, load balancing, etc.?
38. A growing e-commerce company runs its services using separate physical servers: 1 server for the website, 1 server for the database, 1 server for the email system, 1 server for inventory management. The company wants to move to cloud virtualization to improve resource utilization and scaling. Design the cloud virtualization to support multiple services of the company, including facilities of cloud like auto-scaling, load balancing, etc.
39. A mid-sized company runs its payroll system, inventory management, and customer database on separate on-premise servers. During peak business periods, some servers become overloaded while others remain underutilized. The company also faces downtime during maintenance and high hardware upgrade costs. Explain how adopting cloud computing using virtualization and appropriate service models like Infrastructure as a Service or Platform as a Service can solve these issues. Include the role of load balancing, auto-scaling, and resource pooling in your answer.
40. A hospital maintains separate systems for patient records, billing, and diagnostic reports on isolated servers. Due to strict data privacy regulations, the hospital is concerned about security but also wants better scalability

- and availability of services. Analyze how a Private Cloud or Community Cloud can be used to improve efficiency.
41. Explain the role of virtualization, security, and high availability, and suggest suitable cloud service models such as Software as a Service where applicable.
 42. Discuss the benefits of cloud computing architecture.
 43. Analyze how resource pooling impacts regulatory compliance and data sovereignty.
 44. Evaluate the risks associated with rapid elasticity in unpredictable workloads.
 45. Discuss the role of automation in enabling on-demand self-service.
 46. Explain the importance of EDA in cloud computing.
 47. Explain cloud computing architecture with its major components.
 48. Identify the importance of cloud computing in modern IT infrastructure.

Group – C

1. Discuss the role of scalable runtime environments and load balancing in cloud applications.
2. Describe the purpose of load balancers, VPNs, and gateways in IaaS networking.
3. Give two examples of cloud applications. What is Amazon EC2?
4. Explain why SaaS cannot exist without cloud computing infrastructure.
5. How does the automatic scaling impact the performance of cloud platforms?
6. How does Amazon EC2 help developers run applications without managing physical hardware?
7. Explain the importance of firewalls and subnets in IaaS networking.
8. Define Software as a Service (SaaS) and give three real-world examples.
9. Explain the importance of multi-tenancy in SaaS applications.
10. List and explain at least five characteristics of SaaS applications.
11. What are the challenges and risks associated with SaaS adoption in organizations?
12. How does SaaS differ from traditional software installation models?
13. How do cloud services differ from traditional software?
14. Differentiate between middleware and monitoring tools in PaaS.
15. Identify the networking components provided by IaaS.
16. What is Service-Oriented Architecture (SOA), and how does it benefit SaaS applications?
17. Define IDaaS and describe its main functions.
18. Explain how IDaaS improves security and reduces organizational costs.
19. List and explain the core authentication methods supported by enterprise IDaaS solutions.
20. How does SSO improve user experience in IDaaS systems?
21. Describe IaaS with a suitable example.
22. Give examples of PaaS platforms.
23. Explain the term pay-as-you-use in cloud computing with examples.
24. Discuss the impact of regulatory compliance (e.g., GDPR, HIPAA) on cloud providers.
25. What are VM instances in IaaS?
26. Describe the process of how IDaaS verifies and grants access to a user.
27. What are the nine best practices for successful IDaaS implementation? (List any five with explanation)
28. Discuss how cloud services enable collaboration, mobility, and flexible development in modern business environments.
29. Discuss the differences between SaaS and MaaS.
30. Identify the uses of Amazon S3 in cloud computing.
31. Explain the concept of multi-factor authentication and its role in IDaaS.
32. Define the advantages of Open SaaS with suitable examples.
33. Differentiate between cloud services and cloud applications.
34. Discuss the impact of regulatory compliance (e.g., GDPR, HIPAA) on SaaS and IDaaS providers.
35. Explain the importance of centralized identity management in modern cloud environments.
36. Compare Open SaaS and SOA.
37. Write the differences among different types of PaaS.
38. Describe the process of how IDaaS verifies and grants access to a user.

Group – D

1. Explain the concept of data loss in cloud computing.
2. How does authentication differ from authorization in cloud security?
3. What is vendor lock-in?
4. Explain the concept of potential consequences of data loss in cloud computing.

5. Describe User Account Hijacking and suggest preventive measures.
6. What is Vendor Lock-In, and why is it a security concern when changing cloud service providers?
7. Analyze the security implications of Lambda command injection in serverless computing.
8. Differentiate between Role-Based Access Control and Attribute-Based Access Control in cloud security
9. Explain accountability in cloud computing.
10. Discuss Data Security Services in cloud computing, including encryption and backup strategies.
11. How do Network Security Services like VPCs and firewalls protect cloud resources?
12. Describe Application Security Services and their significance in preventing attacks.
13. Explain Monitoring & Threat Detection services in cloud security with examples.
14. What is Cloud Security Posture Management (CSPM), and why is it necessary?
15. Discuss Disaster Recovery & Business Continuity in the context of cloud security.
16. Why is data encryption at rest important for cloud security?
17. How do hackers exploit insecure APIs in cloud environments? Give an example.
18. Identify the major threats to cloud infrastructure.
19. Explain what a Denial of Service (DoS) attack is and how it affects cloud systems.
20. How can shared resources in cloud computing lead to security vulnerabilities?
21. Analyze compliance and legal issues as a security challenge in cloud computing.
22. Explain the threat posed by insider attacks and measures to mitigate it.
23. Your company recently migrated critical applications to a multi-cloud environment (AWS and Azure). A security audit revealed that several storage buckets and databases are publicly accessible, exposing sensitive customer data. How can implementing a CSPM solution help prevent such misconfigurations? What specific actions would it take to improve your cloud security posture?
24. How does DoS attack affect cloud systems? Explain with an example.
25. An organization is migrating critical applications and sensitive customer data to a multi-cloud environment. Evaluate the security and privacy risks associated with cloud computing in this scenario. Propose and justify at least three specific security and privacy mechanisms that should be implemented to protect data, ensure compliance, and maintain confidentiality, integrity, and availability of services.
26. Discuss the risks associated with poor authentication controls in cloud environments.
27. Why is Cloud Security Posture Management necessary?
28. Define different types of Hypervisors in cloud computing.
29. What are the three core information security objectives in cloud computing? Give examples for each.
30. How does authentication differ from authorization in cloud security?
31. Explain accountability in cloud computing and its importance for security.
32. Discuss privacy objectives in cloud security and their real-world applications.
33. Describe isolation in cloud environments and how it prevents interference between tenants.
34. Explain resilience as a cloud security objective and strategies to achieve it.
35. How does ensuring compliance contribute to cloud security? Give examples of standards.
36. How can Server-Side Request Forgery affect cloud infrastructure?
37. Analyze the role of IAM in cloud security.
38. What is Shadow IT?
39. What are open S3 bucket vulnerabilities? Explain their impact on cloud security.
40. Explain the problem of incomplete data deletion in cloud environments.
41. What is Lambda command injection in serverless computing? Explain its security implications.
42. Explain the failure of separation among multiple tenants in cloud computing.
43. What is Server-Side Request Forgery (SSRF)?
44. Explain the concept of lateral movement in cloud attacks and its risks.
45. What is Shadow IT and how does it create security risks in cloud environments?
46. Explain the five essential components of cloud infrastructure security and their functions.
47. Discuss the types of risks in cloud computing, including data breach, vendor security risk, availability risk, and compliance risk.
48. Explain the risk assessment process in cloud computing, including asset identification, threat identification, vulnerability identification, risk analysis, and risk evaluation.

Group – E

1. Describe the steps involved in creating a disaster recovery plan for cloud systems.
2. Analyze the steps involved in implementing internal policy compliance.
3. Differentiate between SSO and multi-factor authentication.
4. Define Internal Policy Compliance in cloud computing.

5. Explain how organizations ensure internal policy compliance in cloud environments.
6. Discuss the importance of security policies in cloud infrastructure.
7. Describe the steps involved in implementing internal policy compliance.
8. Explain the role of access control in enforcing internal policies.
9. Describe the key components of a Disaster Recovery plan.
10. Differentiate between SAML, OAuth, and OpenID.
11. Differentiate between policy compliance and regulatory compliance.
12. Define Disaster Recovery in cloud computing and explain its importance.
13. Explain the working of OAuth in cloud security.
14. What is Recovery Point Objective?
15. Define Regulatory Compliance in cloud computing.
16. Explain the importance of external compliance standards for cloud services.
17. Discuss the challenges of meeting regulatory compliance in cloud environments.
18. Describe the role of government regulations in cloud security.
19. Explain how organizations maintain data privacy under regulatory compliance.
20. Identify the challenges involved in cloud auditing.
21. Explain SAML and its role in authentication.
22. Explain the role of SSL/TLS in securing cloud communication.
23. Explain the concept of Business Continuity Planning in cloud environments.
24. Explain the concept of OpenID and its authentication process.
25. What is the Cloud Security Alliance?
26. Explain how authentication protocols improve cloud security.
27. Differentiate between public cloud, private cloud, and hybrid cloud in terms of security.
28. Design a cloud-based system for an e-commerce platform and explain how you would implement security and privacy mechanisms to protect customer data, including encryption, access control, and compliance with regulatory standards.
29. What is Recovery Time Objective?
30. Differentiate between Business Continuity Planning and Disaster Recovery.
31. Describe the key components of a Disaster Recovery plan.
32. Explain the role of cloud computing in disaster recovery strategies.
33. Discuss different disaster recovery models used in cloud computing.
34. Explain backup strategies for cloud-based disaster recovery.
35. Describe the steps involved in creating a disaster recovery plan for cloud systems.
36. Discuss the advantages of using cloud computing for disaster recovery.
37. How do organizations maintain data privacy under regulatory compliance? Mention methods and tools.
38. Explain Regulatory Compliance in cloud computing with a suitable example.
39. Describe different key management techniques in cloud environments.
40. Write the steps involved in creating a cloud security strategy. Mention methods and tools.
41. Define internal policy compliance in cloud computing.
42. Define cloud audit.